Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software engineer pursuing an MSCS at USC with 2+ years building **scalable backend services**, **REST APIs**, and SQL/NoSQL-backed workflows, with Java/Python alignment, cloud deployments, containerized delivery, and production reliability experience.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Designed and optimized SQL-backed backend services supporting 500+ users, reducing latency from **1.2s to 350ms** through query tuning, indexed access patterns, and scalable service improvements.
- Built ETL-style integration pipelines processing 5,000+ monthly jobs with validation, reconciliation, and CI/CD-backed releases, improving production reliability and accelerating diagnosis of runtime and data issues.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed full-stack React and Flask features handling 10,000+ daily requests, translating product needs into maintainable REST APIs and interfaces while improving usability, testing discipline, and release confidence.
- Implemented secure batch-processing workflows for PDF, DOCX, and ZIP ingestion with AWS-backed persistence, reducing feedback turnaround by **60%** while maintaining stable API response under peak load.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built Python-based evaluation pipelines across 100,000+ records, creating reproducible reporting and debugging artifacts that improved system understanding, implementation quality, and technical communication.
- Produced technical reports and targeted validation scenarios translating findings into actionable guidance, strengthening design-review discussions, reducing handoff ambiguity, and supporting better engineering decisions across teams.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered API and ETL microservices using Python, AWS Glue, and S3, improving response time by **30%** while supporting scalable integrations, structured delivery, and reliable production workflows.
- Maintained 20+ automated test suites using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and improving release confidence across collaborative engineering teams.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a MERN-based platform with WebSockets and REST APIs, supporting 2M+ synchronized records and six workflows through modular service boundaries and responsive user-facing interfaces.
- Automated Docker-based build and test flows with Jenkins, cutting release time by **50%**, improving deployment consistency, and reducing friction during validation and troubleshooting.

Skills

Languages: Java, Python, JavaScript/TypeScript, SQL, C++, Bash
Backend: Spring/Spring Security familiarity, Django, Flask, FastAPI, Node.js, REST APIs, GraphQL concepts, software architecture
Databases: PostgreSQL, MySQL, SQL Server, MongoDB, DynamoDB familiarity, relational schema design, query tuning
Cloud & Delivery: AWS, GCP familiarity, Docker, Kubernetes, Terraform/CDK familiarity, Jenkins, Git pipelines, CI/CD, Angular familiarity

Projects

FinNexus Code | *Next.js, TypeScript, Kubernetes, MongoDB*
- Built a secure fintech platform supporting 50,000+ users and 1M+ transactions, combining real-time updates, scalable data handling, and monitoring patterns for payment-style workflows.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an AI-powered SQL workspace supporting 10+ databases, maintaining Java and TypeScript modules that accelerated schema exploration, query authoring, and structured debugging of data workflows.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed a containerized RESTful data processing API with automated CI/CD releases and validation checks, improving deployment reliability, interoperability, and supportability for production data workflows.